

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ML-POWERED CROP RECOMMENDATIONS FOR FARMERS

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ABSTRACT

Agriculture plays a vital role in the global economy, and selecting suitable crops based on soil and environmental factors is key to sustainable productivity. The **AI-Based Crop Recommendation System** is a machine learning-driven solution that assists farmers in making data-backed crop selection decisions.

The system analyzes parameters such as **temperature, humidity, rainfall, soil type, and pH** to predict optimal crops for a region. Using models like **Decision Trees, Random Forest, and SVM**, trained on historical data, it delivers accurate and reliable recommendations.

Developed with **Python, Scikit-learn, and Flask**, the system provides a **user-friendly interface** for easy access. Future enhancements include integrating **real-time weather data** and **profitability analysis** to further improve precision and promote **sustainable farming**.

INTRODUCTION

Agriculture is a vital sector facing challenges from **climate change**, **soil degradation**, and **unpredictable weather**. Traditional crop planning methods often fail to adapt to these dynamic conditions. To address this, we introduce the **AI-Based Crop Recommendation System**.

This system utilizes **Machine Learning algorithms** to analyze key factors such as **soil type**, **temperature**, **humidity**, **rainfall**, and **pH levels**, providing **data-driven crop suggestions**.

By leveraging models like **Random Forest**, the project delivers **accurate, region-specific recommendations**, promoting **precision farming**, **resource optimization**, and **sustainable agriculture**.

OBJECTIVES AND SCOPE

- Achieve **high predictive accuracy** in recommending suitable crops based on input soil and climatic features using machine learning models.
- **Evaluate and compare** the performance of **Decision Tree, Random Forest, Support Vector Machine (SVM), and Gradient Boosting** algorithms for crop suitability classification.
- **Identify the optimal model** that yields the best performance metrics (e.g., accuracy, precision, recall) on the historical agricultural dataset.
- **Develop a reusable, user-friendly web application** framework (using **Flask**) that seamlessly integrates data input, model prediction, and clear recommendation output.

EXISTING SYSTEM

Existing Solutions:

1. Basic weather/crop advisory apps
2. Govt. portals for schemes & subsidies
3. Local markets for crop selling

Limitations:

1. Generic advice, not personalized Limited AI/ML integration Language & literacy barriers
2. Dependence on middlemen → unfair prices
3. Poor connectivity & offline support gaps

PROPOSED METHODS

- **Multi-Model Approach:** Apply and compare **Decision Tree, Random Forest, SVM, and Gradient Boosting** techniques.
- **Relevant Features:** Utilize soil parameters (N, P, K, pH), **rainfall, temperature, and humidity** data.
- **Data Preprocessing:** Handle missing values, clean data, and **scale/normalize** input features.
- **Model Evaluation:** Assess performance using **Accuracy, Precision, Recall**.

LITERATURE SURVEY

Research Paper / Year	Title / Paper / Work	Methodology / Approach	Inference
“Crop Recommender: Machine Learning–Based”, 2024	[Crop Recommender: Machine Learning–Based Computational Method (conference/paper)][1]	Ensemble and soft-voting classifiers (Naïve Bayes, SVM, DT, RF); uses soil + environmental features; compares individual vs ensemble performance.	Ensemble/soft-voting improved stability and accuracy vs single classifiers — ensembles are recommended for production CRS.
2024 (IJSRA) — Crop Recommendation	[Crop Recommender: Machine Learning–Based (IJSRA / whitepaper)][2]	Uses supervised classifiers (DT, SVM, LR, GNB) and K-Means in preprocessing; input: pH, NPK, temperature, rainfall.	Simple supervised models can reach high accuracy if dataset is clean and balanced; emphasizes that feature choice (NPK, pH, climate) strongly affects results.
M. A. Bakr et al., 2025 (MDPI)	[Evaluation of Learning-Based Models for Crop Recommendation (2025)][3]	Systematic evaluation of 7 ML/DL techniques (including NB, XGBoost) on benchmark datasets; reports accuracy and best models per dataset.	Naïve Bayes & XGBoost achieved top accuracies (~99.5%) on some datasets — shows that tree-boosting and some probabilistic methods remain top contenders depending on data.
M’hamed Mancer et al., 2025	[A Systematic Literature Review of Crop Recommendation Systems for Agriculture 4.0 (CEUR workshop)][4]	Systematic literature review of CRS (2020–2024), covers ML/DL, IoT integration and observed gaps.	Summarizes trends: shift toward IoT+ML, ensemble/boosting models, and a need for uncertainty quantification and farmer-centric UX in deployments. Good source to justify your project scope.
ACM / conference paper, 2025	[Machine Learning-Based Crop Recommendation for IoT-Enabled Precision Farming (ACM proceedings)][5]	Combines IoT streaming sensors with online ML models (incremental learners / tinyML) and evaluates real-time recommendation accuracy.	Demonstrates feasibility of on-device/tinyML crop recommendation with low latency; highlights tradeoffs: edge compute vs model complexity.

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T. Islam et al., 2021	[A Deep Neural Network Approach for Crop Selection and Yield Prediction in Bangladesh][1]	Deep Neural Network (DNN) model compared with SVM, Logistic Regression, and Random Forest using large feature set (weather, soil, fertilizer, etc.).	DNN achieved higher predictive performance for crop selection and yield vs classical ML models on a large multi-feature dataset — shows deep nets help when many input features are available.
FS Prity, 2024	[A Machine Learning Approach to Crop Recommendations][2]	ML pipeline using historical climatic, soil, yield, and farmer-preference data; evaluates classical ML models and ensemble techniques.	ML models can produce useful, personalized crop recommendations; ^{ensembles} and feature engineering improved accuracy and robustness.
P. S. Kiran et al., 2024	[A Machine Learning-Enabled System for Crop ... (MDPI)][3]	Comparative study of ML models (BPNN, SVM, GRNN, MLR) on paddy datasets; uses nutrient + weather features and standard error metrics (MSE, RMSE).	GRNN and certain neural methods showed best performance in their experiments (e.g., high R^2 and low NMSE), indicating non-linear models often outperform linear regression for yield/crop prediction.
M. Bouni et al., 2024	[Integrated IoT Approaches for Crop Recommendation and Precision Agriculture (MDPI)][4]	Integrates IoT sensor streams (soil moisture, temperature, humidity, nutrients) with ML models trained on large sensor datasets; focuses on IoT + ML.	Demonstrates that real-time IoT data combined with ML significantly improves recommendation timeliness and precision; shows scalability with large sensor datasets.
IJARCCCE project, 2024	[Crop Recommendation System Using Deep Learning][5]	Deep learning models (CNN/MLP variants) trained on historical yield, soil, and climatic features; includes data preprocessing and feature selection.	Deep models produced strong classification/regression results for recommending suitable crops; highlights importance of preprocessing and balanced datasets.

METHODOLOGY

Model	Accuracy (TP+TN)	ROC-AUC (Reliability)	Critical Error: (FN)	Conclusion
Logical Regression	0.9181	0.9400	Moderate	Performs well but slightly less accurate; suitable for linear relationships.
Naive Bayes	0.9954	0.8708	Very Low	Very high accuracy, performs best with independent features.
AdaBoost	0.1409	0.7670	High	Poor performance, likely due to unstable base learners
Gradient Boosting	0.9818	0.9322	Low	Strong performance and stable predictions; good generalization.
Random Forest	0.9931	0.9100	Very Low	Most accurate model; chosen as the final model for deployment

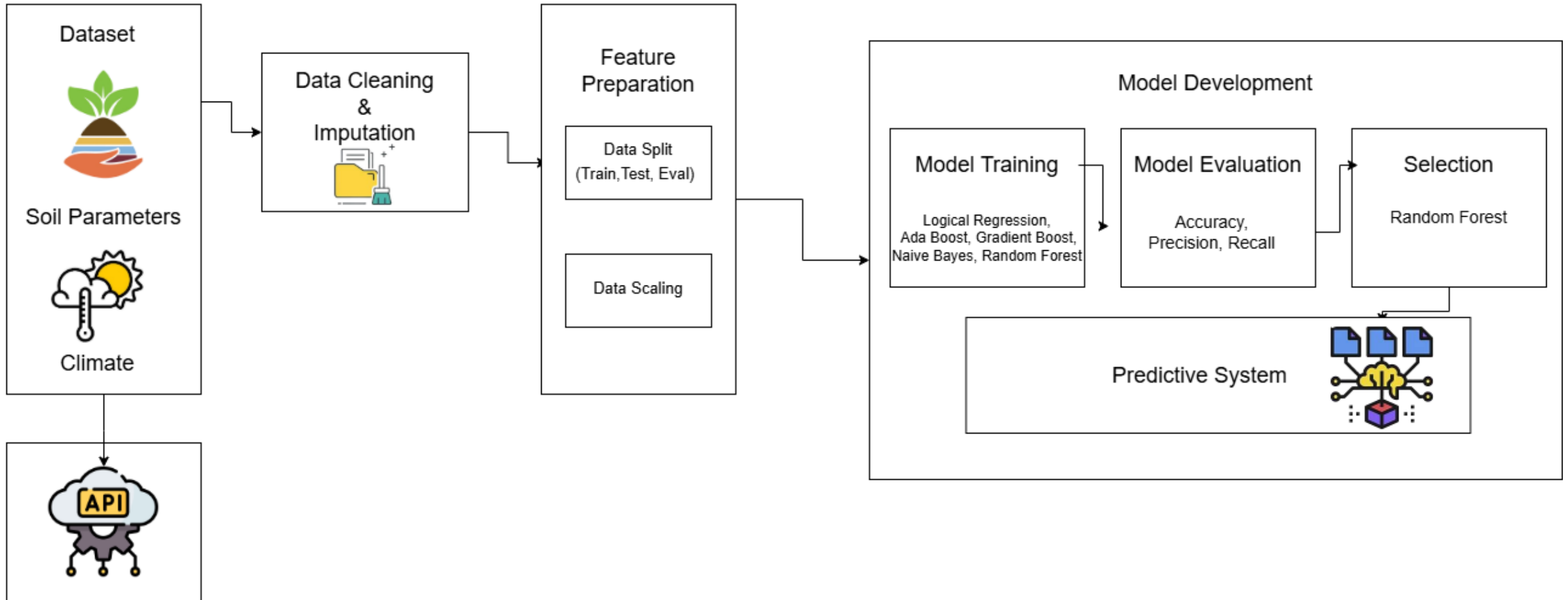
METHODOLOGY

- Collected the dataset from Kaggle, containing features like Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH, and Rainfall.
- Performed data preprocessing by handling missing values, normalizing numerical data, and encoding categorical variables.
- Split the dataset into training (80%) and testing (20%) sets.
- Implemented and compared multiple machine learning models:
 - ✓ Logistic Regression
 - ✓ Naïve Bayes
 - ✓ AdaBoost
 - ✓ Gradient Boosting
 - ✓ Random Forest

METHODOLOGY

- Trained each model on the training data and optimized hyperparameters for better performance.
- Evaluated model accuracy using metrics such as Accuracy Score, Precision, Recall, F1-Score.
- Selected the best-performing model for final crop recommendation.
- Integrated the chosen model into the system to predict the most suitable crop based on input soil and weather parameters.

ARCHITECTURE DIAGRAM



SUSTAINABLE DEVELOPMENT GOALS

- ✓ **SDG 2 (Zero Hunger):** Boost yields through precise crop matching, enhancing food security.
- ✓ **SDG 8 (Decent Work & Growth):** Increase farm profitability by optimizing crop selection and reducing input costs.
- ✓ **SDG 9 (Innovation & Infrastructure):** Drive digital innovation by deploying an AI/ML based decision support system.
- ✓ **SDG 12 (Responsible Consumption):** Optimize resource use (fertilizer, water) by preventing unsuitable crop cultivation.
- ✓ **SDG 13 (Climate Action):** Improve farm resilience and adaptation by matching crops to current temperature and rainfall patterns.

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